

```

clear all;

clc;

vin=30;

D=0.34;

L=500e-6;

C1=100e-6;

C2=50e-6;

C3=100e-6;

C4=100e-6;

C5=100e-6;

R=400;

IL=12.5;

VC1=(2*(1-D)*vin)/(1-2*D);

VC2=vin/(1-2*D);

VC3=(2*vin)/(1-2*D);

VC4=(2*vin)/(1-2*D);

VC5=((3-2*D)*vin)/(1-2*D)

a=[0 0 (2D-1)/(L) 0 0 0; (-1)/C1 0 0 0 0 0; (-2*D+1)/C2 0 0 0 0 0; D/C3 0 0 0 0 (-1+D)/(R*C3); 0 0
0 0 0 -1/(R*C4); (-2*D+1)/C5 0 0 0 0 (-2*D+1)/(R*C5)]

b=[IL*(-2/C2+1/C3-2/C5); 0; VC2*(2/(L)); 0; 0; VC5*(3/(R*C5))]

c=[0 0 0 0 1 1]

d=[0]

ulables=['d'];

ylables=['Vo'];

xlables=['iL vc1 vc2 vc3 vc4 vc5'];

printsys(a,b,c,d,ulables,ylables,xlables)

disp(['Transfer function in s-domain'])

disp(['Vo/d (s)'])

TFb=zpk(tf(ss(a,b(:,1),c,[0])))

```